

AI Based Candidates Recommending System for a Company Vacancy

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Project Proposal Report

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DECLARATION OF CANDIDATE AND SUPERVISOR

We affirm that this proposal is the result of our own work and doesn't include, without proper citation, any material that has been previously submitted for a degree or diploma at any other university or institution of higher learning. To the best of our knowledge and belief, this document does not contain any material authored or published by another individual, except where due acknowledgment has been provided within the text.

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ABSTRACT

Confidence is a critical soft skill in the workplace, which is yet it remains challenging to assess objectively during the process of recruitment. Traditional methods of analyzing confidence in candidates often rely on subjective human judgment, which can be inconsistent and biased. We focus through our research on addressing this issue by developing an innovative AI-driven solution a speech and tone analysis system designed to detect candidates' confidence levels during interviews.

In assessing confident, we have identified the key challenges. They are the variability in patterns of speech, the lack of standardized evaluation methods and the differences in cultures. To overcome these challenges, advanced speech processing and techniques of machine learning to analyze vocal features such as tone, pitch, speech rate, and emotional cues are leveraged by our system. To classify confidence levels in real-time, providing recruiters with objective and data-driven insights, these features are used.

This confidence module detection into a web-based AI candidate recommendation system integrated by our solution. Other than the confidence analysis, the system evaluates other soft skills and qualifications such as stress level, decision making skill and the engagement skill of candidate offering a comprehensive assessment of each of them. We hope to transform the hiring process, by combining technology of cutting-edge with practical recruitment tools, making it more reasonable, efficient, and effective.

Empower companies to make better hiring decisions by providing them with a reliable and scalable tool for assessing confident of candidate is our goal. We hope that this innovation not only to develop the process of recruitment but also contribute to build more confident and improve the capability of teams in the workplace.

Keywords: Confidence detection, speech analysis, tone analysis, AI recruitment, soft skills assessment, machine learning, web application, candidate recommendation system.

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1. INTRODUCTION

Assessing candidates objectively is important to ensure for the selection of the most suitable individuals, in the competitive landscape of recruitment. Yet traditional methods lack objectivity and scalability, confidence levels significantly influence interview performance. This proposal report depicts an innovative AI-based approach utilizing speech and tone analysis to predict confidence levels during interviews, thereby addressing biases inherent in evaluation of humans.

1.1 Background

Confidence is a critical soft talent that plays an important role, in discovering a success of a candidate in the workplace. It provides an impact on the skills such as communication, decision-making, and the ability to face challenges in proper way. However, for employees during the process of recruitment assessing confidence remains a significant challenge. Traditional methods of assessing confidence, like interviews and subjective observations are frequently inconsistent, biased and time consuming. These methods mostly rely on human judgment, which can differ in a great area depending on the interviewer's experience, perspective and personal prejudices.

In today's competitive job market, companies quickly utilizing AI-driven solutions to streamline and enhance their process of recruitment. Huge data amounts can be examined by AI systems objectively, providing insights that are difficult to achieve through manual approaches. However, assessing soft qualities such as confidence still remains an area to develop, while AI has made great progress in evaluating technical skills and qualifications.

Speech and tone analysis offers a potential remedy to overcome this challenge. It shows the vocal features such as pitch, tone, speech rate and emotional cues have a close relationship for the confident level of an individual by research in psychology and computational linguistics. For instance, a steady tone, moderate speech tempo and clear articulation are often associated with high confidence, while hesitations, monotone speech or excessive pauses may be indicative of poor confidence. By leveraging cutting-edge machine learning algorithms and technologies of speech processing, it can evaluate these vocal characteristics and offer objective, data-driven evaluations of a candidate's confidence.

Although there's a potential, implementing speech and tone analysis to detect confidence is nothing without challenges. Differences in patterns of speech because of accents, variations in cultures and emotional states can make analysis complex. Additionally, ensuring the ethical use technology like that avoiding biases and protecting privacy of a candidate is critical to gain success of it.

We intend to pay attention on our research to call these challenges by developing a strong and dependable system to detect confidence from the analysis of speech and tone. We target to supply recruiters with a strong tool for evaluating soft skills with objective and efficiently, by integrating this system into a web-based platform which recommend candidates through AI. This invention has the ability to convert the process of recruitment, enabling companies to get out confident and candidates with capabilities who are well-suited to the necessities of the hiring company.

1.2 Literature review

Recently, the assessment of soft skills especially confidence has generated a lot of buzz as it is key to job success. Traditional methods for measuring confidence such as subjective reports and interviews are often too unreliable and biased. In an attempt to surmount these limitations, researchers have explored new niche technologies that use voice and tone analysis to measure and quantify even objectively confidence.

1. Analyze Speech and Tone in Psychology and AI

The monitoring of patterns in speech and vocal tone has been an established research field in psychology to understand human emotional status and patterns in behaviors. through research, Scherer et al. (2003) depicted how vocal elements consists with pitch and tone along with speaking speed exhibit strong relationship with both emotional status and confident level of an individual [1]. To build models which can be used to analyze speech patterns to determine psychological traits, AI-driven systems are applied these research findings. Schuller et al. (2011) invented a machine learning framework to identify emotions in speech which shows capability of AI in psychological evaluation through analysis of vocal signal.

2. Deep Learning and Machine Learning in Speech Analysis

Improvements in deep learning and machine learning offer significantly better accuracy for voice and tone analysis at present. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are functioned as popular deep learning models to process speech data and to monitor tasks. Trigeorgis and colleagues (2016) created a deep learning model which processed raw audio to achieve state-of-the-art performance in end-to-end speech emotion identification through complex pattern extraction [3]. According to Satt et al. (2017), deep learning models successfully detect confidence levels from speech through effective feature extraction and precise model architecture which leads to high accuracy [4].

3. Recruitment Applications and Candidate Assessment

Now, Recruitment systems include speech and tone analysis to develop evaluation methods of candidate. Nguyen and colleagues (2019) designed an artificial intelligence-powered recruitment system that analyzes speech to assess confidence of candidates and communication skills representing how this technology can be helped to minimize bias and enhance results of hiring [5]. A multi-modal AI system which merges speech analysis with expressions of face recognition to evaluate soft skills of a candidate demonstrating AI's adaptability in recruitment processes [6] was developed by Kumar et al. (2020)

4. Limitations and Challenges

Analysis of Speech and tone to detect confident depicts promise but encounters multiple obstacles. The analysis becomes complex when speech patterns show variability because of accents along with cultural differences and emotional states. Li et al. (2018) pointed out the requirement for robust AI models capable of generalizing across different populations and contexts [7]. Privacy and bias focus a resolution for demand to ensure that these technologies operate responsibly. The research by Smith et al. (2021) highlighted the more need for transparent and fair practices in AI recruitment tools which needs strict validation and testing to avoid bias issues [8].

5. Integration with Web-Based Recruitment Systems

Recent research has concentrated on merging speech and tone analysis capabilities into web-based recruitment systems. Patel and his team presented in 2020 a real-time speech analysis tool for a web application in recruitment that shows its capability to deliver immediate feedback about candidate confidence levels [9]. A system for cloud computing which merges speech analysis with AI-powered tools consisting resume parsing and skill matching to deliver whole evaluation of candidate [10] was introduced by Zhang et al. (2021)

6. Directions for the Future

Speech and tone analysis improvements for recruitment will need systems for becoming more adaptive to variant contexts. Wang et al. (2022) conducted research on transformer-based models including BERT and Whisper which showed exceptional results in speech analysis by detecting subtle vocal indicators of confidence [11]. Confidence detection systems will attain higher accuracy when multi-modal data including facial expressions and body language become part of their analysis [12].

1.3 Research Gap

While speech and tone analysis is a good area to go through in the research of assessing confident, Several gaps do exist in the prevailing methodologies and the technologies. In here report, we focus on the key aspects of detecting the confidence and highlight the limitations of prevailing approaches, supported by relevant studies.

1. Confidence Level Prediction

Prediction of the confident level will need to analyze characteristics of voice like tone, pitch and speech rate to infer the confidence of a speaker. The feasibility of utilizing machine learning to predict confidence and doubt in accented speakers, their work center focuses on human perception rather than automated systems, have been represented by the studies like Rojas and Bilmes [14]. Lack of strong, generalizable models exist which can be used to predict confidence levels across variant populations and contexts accurately.

Research Gap: Struggling current systems with variability in patterns of speech due to accents, differences in cultures, and emotional status. A require for adaptive models exist that can be managed these variations when maintaining quality accuracy.

2. Speech Analysis in Real-time

For applications which are recruitment, real-time speech analysis is distinct, where quick feedback on confidence of a candidate is important. For extracting real-time acoustic features, OpenSMILE [13] and Praat [15] are tools that supply powerful frameworks. However, those tools are frequently used to control environments and may not be performed good in real-world settings or noisy.

Research Gap: Prevailing real-time speech analysis systems aren't optimized for environments with noise or scalable integration into web-based platforms in recruitment

3. Acoustic Feature Extraction

Extraction acoustic feature is the pillar at base of confidence detection. OpenSMILE [13] and Praat [15] which considered as tools offer comprehensive feature extraction capabilities. Pitch, spectral features and intensity are included. However, it's still a challenge which the selection of relevant features for confidence detection. The importance of feature selection in speech identification in emotion, but this wasn't fully explored for confidence detection was focused by Schuller et al. [16].

Research Gap: No proper standardized feature sets are there and methodologies to extract acoustic features based on confidence.

4. Integration with Recruitment Systems

Standalone detection of confidence systems had been built during, there is limited research on integrating these systems into platforms of AI-driven recruitment. Nguyen et al. [17] explored the benefits of speech monitoring in recruitment but their work fails to address the integration challenges of real-time, user experience and scalability in web-based applications.

Research Gap: At present systems are not designed for seamless integration with platforms in recruitment, making limits to their practical applicability.

5. Speech Emotion Recognition

Speech emotion recognition (SER) is associated closely to detect confidence, as both rely on analyzing features of voice. A SER comprehensive review, concerns on its challenges and potential was given by Schuller et al. [16]. However, SER systems are typically created to detect broad emotional states (happiness, anger) rather than special traits such that confidence.

Research Gap: Rather than treating it as a byproduct of recognition of emotion, need for specialized models that direct to detect confidence as a distinct trait is there

Table 1: Research Gap

Aspects	Limitations at present	Proposed Solution
Confident level prediction	Limited generalizability across accents, cultures, and emotional states.	Develop adaptive models using diverse datasets and transfer learning techniques.
Real time speech analysis	Poor performance in noisy environments; lack of scalability for web-based platforms	Optimize algorithms for noise robustness and real-time processing.
Extraction of Acoustic Feature	Standardized feature sets for confidence detection lack	Recognize and validate confidence-specific acoustic features.

Integration with Recruitment Systems	Real-time research limitations, scalable integration.	Design modular systems for seamless integration into platforms in recruitment.
Recognition of Speech Emotion	Concerns on broad emotions rather than specific traits like confidence.	Develop specialized models for confidence detection.

Conclusion

The aim of the proposed system is to call the gaps in here by integrating machine learning techniques with adaptability in region specific and considerations in ethical in advance. By developing a robust, scalable, and fair system in confidence detection, we target to improve the accuracy and reliability of AI-driven platforms in recruitment, by making them more effective and inclusive.

1.4 Research Problem

The traditional selection of candidates for vacancies in company rarely assesses specific soft skills, such as confidence, stress level detection, decision making skill and engagement skill. On resumes, technical assessments, and subjective interviews these methods are primarily included, which provide limited insights into a candidate's ability to thrive in a dynamic workplace. Result of this, mismatches between job roles and candidates have to be faced by the organizations, leading to raised turnover rates, productivity reduction, and inefficiencies in the process of hiring.

Confidence is one of the major soft skills, which is again influenced by the candidate's art of communication, decision-making, and handling challenges. Confidence is highly subjective to assess through traditional methods. This biasedly sets the thinking of the interviewers, who unconsciously are likely to like those who present extroverted behaviors or who share their backgrounds while losing other better candidates who can also offer substantial results, although more introverted. Subjectivity does not equate to fair hiring, which may affect an organization in making proper judgments that balance technical competence with interpersonal ability.

This becomes a promising problem due to the recent enhancement in artificial intelligence and speech analysis. In using AI-driven tools, this will be capable of objectively analyzing the speech and tone of a candidate to perceive the confidence level. However, present systems that detect confidence are burdened by issues like the following:

1. Generalizability: Most of the systems are trained with homogeneous datasets, thus making their efficiency poor under diversified cultural and linguistic contexts.
2. Real-time Processing: Current tools often do not consider real-time processing, crucial in real integration at the point of recruitment.
3. Integration Problem: Seldom have the difficulties of integrating a system of confidence detection into any existing recruitment workflow been examined
4. Ethical Considerations: AI is used in recruitment to raise number of considerations about privacy and transparency.

This research targets to fill these gaps by developing a system of AI-powered that can detect confidence using speech and tone monitoring, integrated with a web-based candidate recommendation platform.

The paper goes beyond the conventional approach of utilizing modern technologies to help organizations in conducting a more objective, efficient and inclusive soft skills assessment process while hiring.

2. OBJECTIVES

2.1 Main Objective

For building a system which is AI-driven that predicts the confidence level of candidate through speech and tone analysis during the process of interviews is the main objective of this research. Data-driven insights into a candidate's soft skills, increase the accuracy and fairness of the process of recruitment will be provided the system.

2.2 Specific Objectives

These are a handful of the sub objectives which fulfill the primary objective.

1. Determine Primary Acoustic Elements

- Examine and identify the auditory features (pitch, speech rate, intensity, modulation) that that best reflect the confidence levels in speech.
- Validate the relevance of these characteristics through empirical analysis and comparison with psychological studies on voice expression.

2. Create Models for Machine Learning

- Create and train models for machine learning to analyze speech data and categorize confidence levels (low, medium, high).
- To increase the strongness and accuracy of the system, explore built techniques like deep learning (RNNs, CNNs) and pre-trained models (Wav2Vec, Whisper).

3. Integrate with Web Application For the interview

- Construct a modular and scalable system that combines the speech analysis model with a web-related interview platform.
- By giving recruiters with actionable insights, while in interviews, enable real-time feedback on candidates' confidence levels

4. Assess System Performance

- Compare the predictions of the system with human evaluations to confirm its effectiveness and identify areas for improvement.
- Assess the performance of the system across various datasets

3. METHODOLOGY

The suggested remedy entails building an AI-driven system that analyzes speech and tone during the process of interviews, for predicting the confidence level of a candidate. The below major steps are defined the methodology of the proposed solution.

1. Audio Source Input

- Gather Data:
 - Compile auditory recordings of candidate interviews which accessible through datasets in the public
 - Based on human assessments, datasets are marked with confidence levels as low, medium and high.
- Preprocessing:
 - For consistency, audio files are transformed into a standardized format (WAV).
 - Using noise reduction algorithms, background noise is removed and improve quality of audio
 - For additional processing, audios are segmented into smaller chunks (5-10 seconds).

2. Recognize Features of Voice

- Extraction of Feature:
 - Open sources tools like openSMILE [1] or Praat [2] are used to extract acoustic features from the audio data.
 - Concerns on features such that:
 - Pitch: Variations in Vocal tone.
 - Intensity: Speech volume or Loudness.
 - Speech Rate: Number of words that speaks in a minute
 - Pauses: Frequency and Silences duration.
 - Modulation: Intensity over time and changes in pitch.
 - More characteristics like spectral characteristics (Mel-frequency cepstral coefficients - MFCCs) and emotional cues (sentiment analysis) can be extracted.
- Selection of the feature:
 - To find the most matching features for confidence detection, use statistical methods (correlation analysis) and machine learning techniques (feature importance ranking)

3. Analyze Voice Features

- **Model Development:**
 - Train machine learning models (Support Vector Machines, Random Forests) and deep learning models (CNNs, RNNs, Transformers) to divide confidence levels based on the extracted features.
 - With limited data, to build performance, utilize models which trained previously (Whisper, Wav2Vec) for transfer learning.
 - To connect the strengths of multiple models for better accuracy, implement ensemble techniques
- **Training of Model and Validation:**
 - Datasets are split into training, validation, and test sets (70% training, 15% validation, 15% testing).
 - Using the validation set, train the models on the training set and tune hyperparameters
 - Using metrics such as accuracy, precision, recall, and F1-score, evaluate the models on the test set

4. Predict Confidence Level

- **Real-Time Analysis:**
 - Integrate the trained model into a real-time speech processing pipeline.
 - Use streaming audio input from candidate interviews to extract features and predict confidence levels on-the-fly.
 - Provide quick feedback to recruiters through a user-friendly interface.
- **Confidence Level Classification:**
 - Classify confidence levels into categories (low, medium, high) based on predefined thresholds or probabilistic outputs from the model.
 - Include a confidence score (0-100%) to provide a more nuanced assessment.

5. Notify the Company

- **Integration with Recruitment Platform:**
 - Build a web-based application that integrates the confidence detection system with existing recruitment platforms.
 - Use APIs to send real-time notifications to recruiters, displaying the candidate's confidence level and supporting insights (key vocal features).
- **User Interface Design:**
 - Design an intuitive dashboard for recruiters to view candidate confidence levels, compare multiple candidates, and access detailed reports.
 - Include visualizations (graphs, heatmaps) to highlight trends and patterns in candidate performance.

Tools and Technologies:

- Speech Processing: openSMILE [1], Praat [2], Librosa, PyAudio.
- Machine Learning Frameworks: TensorFlow, PyTorch, Scikit-learn.
- Pre-trained Models: Whisper, Wav2Vec.
- Web Development: Flask, Django, React.
- APIs: Google Cloud Speech-to-Text, IBM Watson, Azure Speech Services.

This is the overview of the Confident detecting system which can predict the confident level of candidates during the interviews. Also, it is depicted the overall system which connected all components to select the best candidate for the relevant job role.

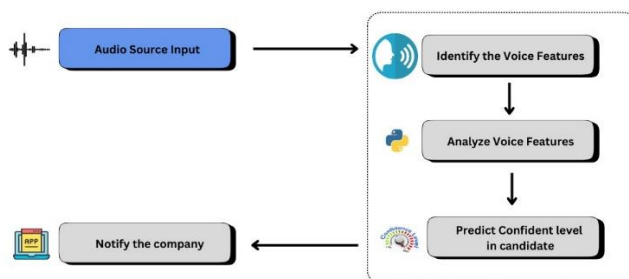


Figure 3. 1 Engagement Analysis component overview diagram

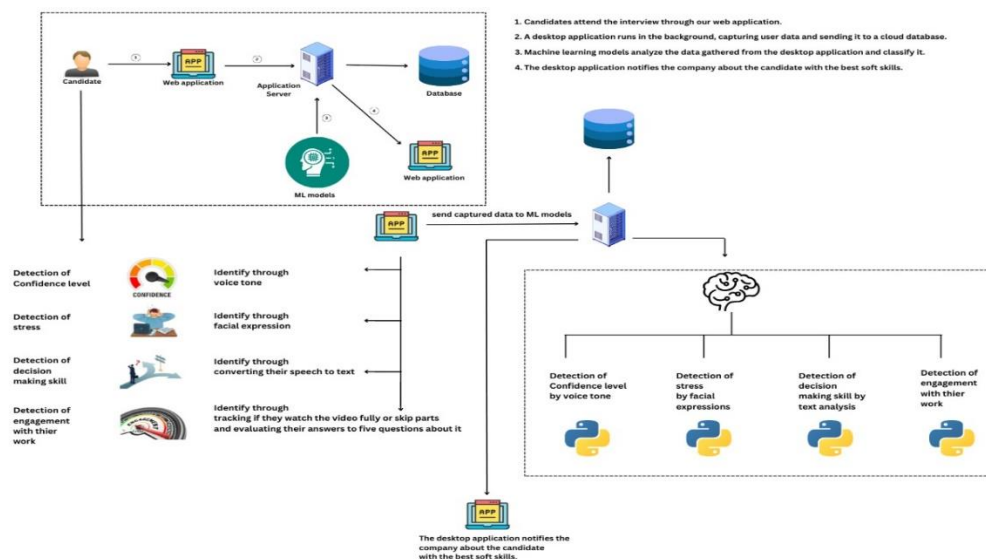


Figure 3. 2 overview System diagram

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

To achieve the goals, these are some functional requirements that must be met.

1. Capture the Audio Input:

- While interviewing the candidate, the capability to record audio in high quality input
- Support numerous auditory formats (WAV, MP3) and confirm compatibility with different recording devices like microphone.

2. Speech Analysis:

- Speech characteristics are examined to be measured include pitch, tone speech rate, pauses and intensity.
- Associated auditory features using advanced speech processing tools (openSMILE, Praat) are extracted

3. Predict the Confidence Score:

- Confident level of the candidate is estimated based on examined speech characteristics
- Confidence score will be in the range of 0-100% and categorize confidence into levels as low, medium and high

4. Real-Time Feedback: Expect the real time feedback

5. For Recruiters dashboard should be there:

- The system must display results in an intuitive and user-friendly dashboard for recruiters.
- Graphs and confident scores are included as visuals for comfortable interpretation
- For the comparison of multiple candidates and for making detailed reports, recruiters are allowed

4.2 Non-Functional Requirements

To ensure the reliability, scalability and security the system must meet the following non-functional requirements

1. Processing in Real-Time:

- The system must process audio data in real-time with minimal latency (less than 1 second delay).
- Use infrastructure and algorithms for rendering their potential to handle effectively

2. Privacy and Data security

- Encryption protocols are used to guarantee the security and privacy of captured speech samples
- Offer facilitates for candidates to agree for auditory recording and analysis.

3. Scalability

- Ability to handle the system with multiple users at the same time without having degradation
- Prioritizing demand, cloud-based infrastructure as AWS and Google Cloud) should be used to support dynamic scaling.

4. Accuracy and Reliability

- The system must gain high accuracy (>90%) in the estimating process confidence levels, validated over human evaluations
- Ensure the robustness across diverse datasets, concerning variations in accents, languages and cultural contexts

5. Experience of User

- The platform should be intuitive and user-friendly, providing a seamless experience for both recruiters and candidates.
- It must be compatible across different devices, including desktops, laptops, and tablets, and support various browsers like Chrome, Firefox, and Safari

6. Ability to maintain

- This should be designed for the architecture of Modular, to make it easy for maintain and updates.
- Continuous integration/continuous deployment (CI/CD) pipelines and version control systems (Git) should be utilized, to support efficient development and deployment.

4.3 User Requirements

The user requirements are classified into functional, non-functional and usability requirements to ensure the requirements of all stakeholders

	Functional	Non-functional	Usability
Recruiters	-The system should extract the key features -System should predict the confident level -System must provide notifications of the confident level -The system must handle multiple candidates	-System must process and provide real-time feedback -Able to work continuously -Achieve $\geq 90\%$ accuracy in the prediction of the accuracy	-Should provide clear instructions and feedbacks -System must be easy to set up, with a setup time of ≤ 10 minutes
Candidates	-The system able to capture high-quality audio input during the interview (To make comfortable) -System must filter out background noise and process speech accurately	-	-System should be comfortable to use without making distracts
Administrator	-Store and retrieve data securely -Integrate with existing recruitment platforms via APIs	-System must be compatible with operating systems -Encrypt all audio recordings and candidate data -System must comply with data protection regulations	-

User Stories

Recruiter:

- As a recruiter, I want to capture high-quality audio during interviews so that I can accurately assess candidate confidence.
- As a recruiter, I want real-time notifications about candidate confidence levels so that I can make informed decisions during interviews.

Candidate:

- As a candidate, I want the system to be non-intrusive, in order to focus on my interview without interruptions.
- As a candidate, I want my data to be secure and private so that I feel comfortable using the system.

Administrator:

- As an administrator, I want the system to integrate with existing recruitment platforms so that I can streamline the hiring process.
- As an administrator, I want the system to comply with data protection regulations so that I can ensure legal and ethical use.

4.4 System Requirements

Hardware Components

1. Microphones with high-quality:

- To capture audios clearly and accurately we should have to use qualified microphones when the interview is going on

2. Unit of Processing:

- To manage the real-time speech processing and analyzing it is needed a high-performance processing unit
- It is needed Cloud-based infrastructure like AWS, Google Cloud for scalable and efficient processing.

3. Storage:

- Enough capacity for storage is needed to keep the auditory recordings, model outputs and extracted features
- Solutions of Cloud storage can be utilized for scalability and accessibility.

Software Components

1. Speech Processing Libraries:

- **openSMILE**: For extracting acoustic features such as pitch, intensity, and speech rate.
- For analyzing speech and extracting related features, **Librosa** is used for audio analysis, while **Praat** helps to extract special speech parameters and to perform phonetic analysis.

2. Machine Learning Frameworks:

- For building and training models, **TensorFlow and PyTorch** are used to develop deep learning architectures like CNNs and RNNs for detecting confidence in speech.
- **Scikit-learn** is used for the implementation of traditional machine learning models such as SVM and Random Forest, which act as baseline models for comparison with more advanced techniques.

3. Pre-trained Models:

- To improve the accuracy of speech analysis, **Whisper (OpenAI)** is used to convert speech to text and to extract primary characteristics.

4. Web Development Frameworks:

- The backend of the application is built using **Flask or Django**, depending on the project's requirements.
- On the frontend, **React or Angular** are used to create a smooth and interactive user experience.

5. Speech Processing APIs:

- To enable real-time speech recognition and transcription, **Google Cloud Speech-to-Text** is integrated.
- **IBM Watson** Speech-to-Text is used as an alternative for improving flexibility and accuracy.

6. Management of Database:

- Analysis results and candidate details are stored in **MySQL or PostgreSQL** for structured data.
- **MongoDB** is used to maintain unstructured data like audio recordings and extracted features.

4.5 Expected Test Cases

To ensure the system operations as intended, a comprehensive set of test cases must be designed to validate performance of system, accuracy and usability. Expected test cases are classified into different types as functional, non-functional, usability and edge case scenarios

Functional Test Cases

Test case ID	Description	Input	Expected Output	Pass/ Fail Criteria
FTC-01	Verify real-time audio capture using a microphone	Candidate speaks during the interview	System captures audio and process speech in that time	Audio is recorded without a lag
FTC-02	Detection of key auditory features	Candidates speak with pitch, intensity etc.	System detects and extracts those features	Identify auditory features correctly on real-time
FTC-03	Test real-time notification for recruiters	Candidate exhibit low and high confidences during the interview	Recruiter receives a notification on different confident levels	Notification is send to recruiters when changing the levels of confident

Non-Functional Test Cases

Test case ID	Description	Input	Expected Output	Pass/ Fail Criteria
NFTC-01	Verify the time taken to respond the system for real-time	Candidate speaks during the interview	System predicts confident level within 2 seconds	Confident level is predicted within a specific time
NFTC-02	Scalable the system to use multiple users	5-10 candidates take part in the interview	System provide performance well to candidates without any issues	No lag when there is multiple users
NFTC-03	System reliable on over usage	System works continuously in working hours (8 hours)	System maintains the consistency	No system failure during test period

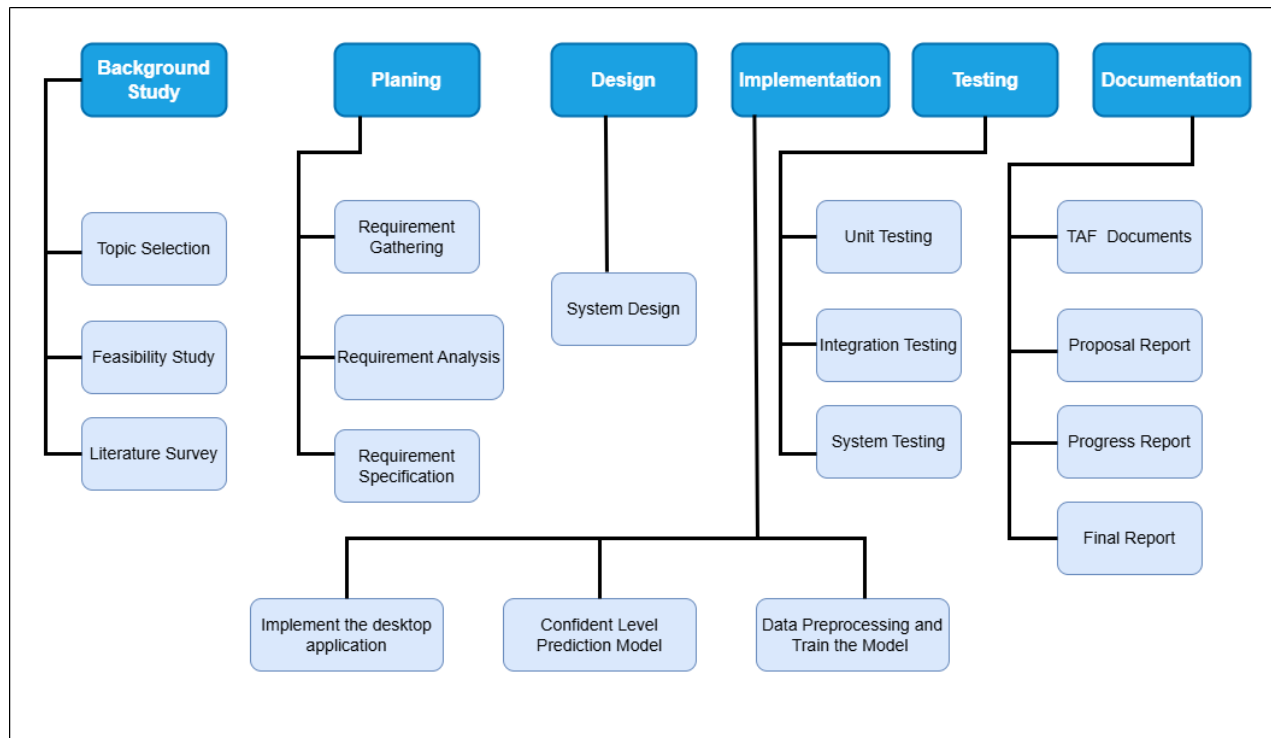
Usability Test Cases

Test case ID	Description	Input	Expected Output	Pass/ Fail Criteria
UTC-01	Test the experience of candidate during interview	Candidate face the interview vis system	Candidate doesn't have any uncomfortable	Feedback of the candidate is non-intrusive and seamless experience
UTC-02	Test ease of setup for recruiters	Recruiters set up the system for the interview	System can be used within 10 minutes	System can be set without technical assistance

Edge Case Test Cases

Test case ID	Description	Input	Expected Output	Pass/ Fail Criteria
ETC-01	Test the behavior of the system without speech input	Candidate remains silent for a period	System detects absence of speech and logs it appropriately	System handles the scenario without errors
ETC-02	Verify system with extreme behavior patterns	Candidate speaks very fast or very slow	System accurately distinguishes between normal and extreme speech patterns	Confident level detection is accurate although the extreme speech patterns changed

5.WORK BREAKDOWN STRUCTURE



6.GANTT CHART

	2024	2025											
Tasks	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Project Initiation													
Finding the topic													
Finding the Client													
Finding a Registration													
Project Planning													
Identifying the Problem													
Feasibility Study													
TAF Documentation													
TAF Evaluation													
Project Proposal Presentation Preparation													
Project Proposal Evaluation													
Project Proposal Report Documentation													
Project Proposal Report Submission													
Implementation													
Collecting Data													
Component 1													
Component 2													
Component 3													
Component 4													
Progress Presentation													
Integration													
Research Paper Publication													
Testing													
Unit Testing													
Component Testing													
Integration Testing													
Progress Presentation 2													
Deployment													
Deployment to Production													
Final Report Presentation													
Final Presentation and Viva													
Final Report Submission													

7.BUDGET & Budget Justification

A thorough financial breakdown for the creation and implementation of the suggested real-time confidence detection system for interview applicants may be found below. The budget covers staff, software, hardware, and other related expenses. Every item is supported by the fact that it is essential to the project's successful completion.

7.1 Budget Description

Category	Item	Estimated Cost (LKR)	Justification
Hardware	Qualified Microphone	20000	Required for capturing clear facial expressions during interviews.
	Cloud computing resources	10000	Needs to scalable and efficient real-time speech processing.
Software	Speech Processing Tools (openSMILE, Librosa)	0	Open-source tools to vocal feature extraction.
	Machine learning frameworks (TensorFlow, PyTorch)	0	Open-source frameworks for developing and training the confident detection model.
	Web Development frameworks (Flask, React)	0	Open-source frameworks to build the user interface.
Dataset	Available on internet freely can be used	0	To train the machine learning model
Miscellaneous	Contingency (10% of total budget)	3000	Buffer to unforeseen expenses
Total Estimated Budget		33000	

7.2 Budget Justification

- Expenses for Hardware:
 - A qualified microphone is needed to record crystal-clear auditory input and guarantee precise extraction of feature.
 - Scalable and effective real-time voice processing, particularly when maintaining several concurrent interviews, requires resources of cloud computing.
- Allowance for the software:
 - Open-source tools Costs are minimized while using reliable and well supported technology with the help of tools such as openSMILE, TensorFlow, and Flask.
- Datasets are essential for teaching the machine learning model to forecast confidence levels with accuracy. These can be taken free through online resources.
- Unspecified Fees:
 - Contingency fund: Added to ensure the project can adjust to unforeseen obstacles by covering unforeseen costs.

8. REFERENCES

- [1] “APA PsycNet,” *psycnet.apa.org*. <https://psycnet.apa.org/record/2009-07773-023>.
- [2] B. Schuller, S. Steidl, and A. Batliner, “The INTERSPEECH 2011 Speaker State Challenge,” *Interspeech*, 2011.
- [3] G. Trigeorgis, F. Ringeval, and R. Brueckner, “End-to-end speech emotion recognition using deep neural networks,” *IEEE Transactions on Affective Computing*, 2016.
- [4] A. Satt, S. Rozenberg, and R. Hoory, “Efficient Emotion Recognition from Speech Using Deep Learning on Spectrograms,” *Interspeech 2017*, Aug. 2017, doi: <https://doi.org/10.21437/interspeech.2017-200>.
- [5] T. Nguyen, H. Nguyen, and T. Tran, “AI-driven recruitment: A case study on speech analysis for candidate assessment,” *Journal of AI in HR*, 2019.
- [6] S. Kumar, A. Sharma, and P. Gupta, “Multi-modal AI for soft skills assessment in recruitment,” *IEEE Transactions on Human-Machine Systems*, 2020.
- [7] X. Li, Y. Zhang, and Z. Chen, “Robust speech emotion recognition across diverse populations,” *IEEE Transactions on Affective Computing*, 2018.
- [8] J. Smith, T. Brown, and R. Davis, “Ethical considerations in AI-driven recruitment systems,” *Journal of AI Ethics*, 2021.
- [9] R. Patel, A. Singh, and V. Kumar, “Real-time speech analysis for web-based recruitment platforms,” *International Conference on Web Applications*, 2020.
- [10] L. Zhang, Y. Wang, and H. Liu, “Cloud-based AI recruitment systems: A comprehensive approach,” *IEEE Cloud Computing*, 2021.
- [11] X. Wang, Y. Chen, and Z. Li, “Transformer-based models for speech analysis in recruitment,” *IEEE Transactions on Neural Networks*, 2022.
- [12] A. Gupta, P. Sharma, and R. Kumar, “Multi-modal AI for candidate assessment: Integrating speech, facial, and body language analysis,” *Journal of AI in Recruitment*, 2022.
- [13] F. Eyben, M. Wöllmer, and B. Schuller, “openSMILE – The Munich Versatile and Fast Open-Source Audio Feature Extractor,” in *Proc. 18th ACM Int. Conf. Multimedia*, 2010, pp. 1459–1462. doi: [10.1145/1873951.1874246](https://doi.org/10.1145/1873951.1874246).
- [14] J. D. Rojas and J. A. Bilmes, “Predicting Confidence and Doubt in Accented Speakers: Human Perception and Machine Learning Experiments,” in *Proc. Interspeech 2016*, 2016, pp. 2150–2154. doi: [10.21437/Interspeech.2016-1426](https://doi.org/10.21437/Interspeech.2016-1426).
- [15] P. Boersma and D. Weenink, “Praat: Doing Phonetics by Computer,” Available: <http://www.praat.org/>.
- [16] B. Schuller *et al.*, “Speech Emotion Recognition: State of the Art and Remaining Challenges,” *Cognitive Computation*, vol. 10, no. 4, pp. 1–19, 2018. doi: [10.1007/s12559-018-9522-2](https://doi.org/10.1007/s12559-018-9522-2).
- [17] T. Nguyen, H. Nguyen, and T. Tran, “AI-driven recruitment: A case study on speech analysis for candidate assessment,” *Journal of AI in HR*, 2019.

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